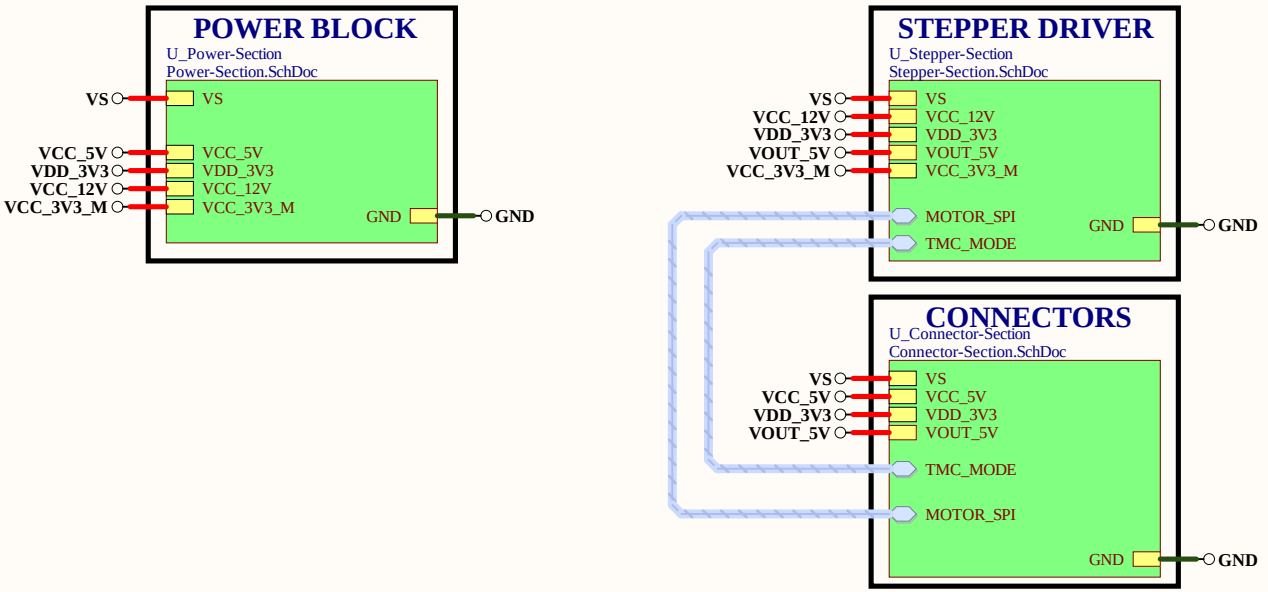
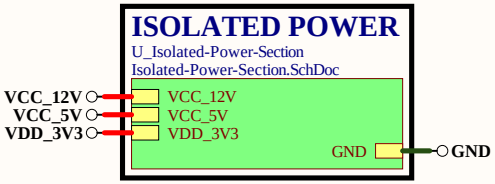
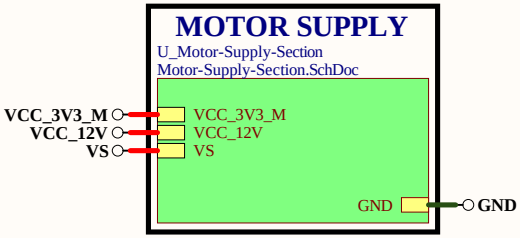
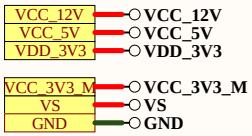
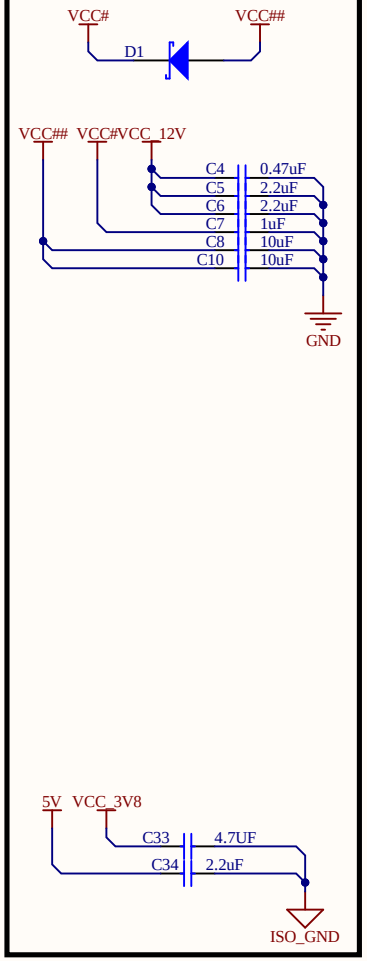

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
MOTOR_SPI
TMC_MODE
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
VCC_12V
VCC_5V
VDD_3V3
GND
GND

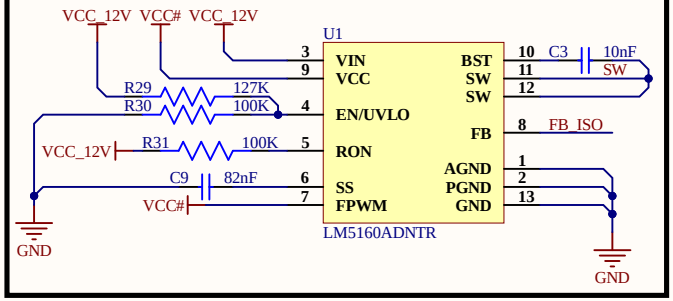

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VCC_3V3_M
VCC_12V
VS
VCC_3V3_M
VCC_12V
VS
GND
GND


VCC_12V
VCC_5V
VDD_3V3
VCC_12V
VCC_5V
VDD_3V3
VCC_3V3_M
VS
GND
VCC_3V3_M
VS
GND

DECOUPLING CAPS

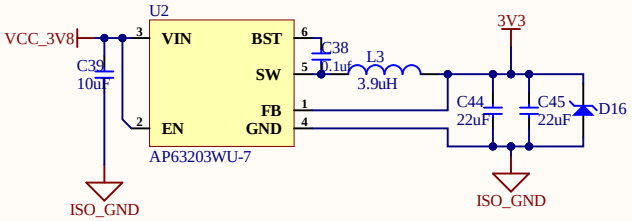


12V-3.8V ISOLATED BUCK CONVERTER

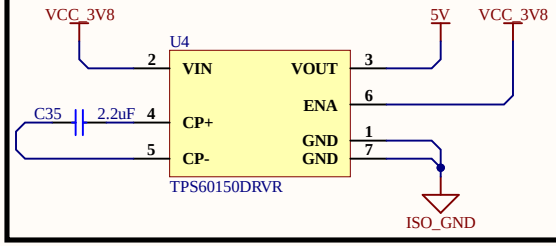


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

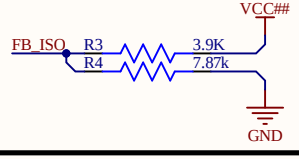
3.8-3.3V LINEAR REGULATOR



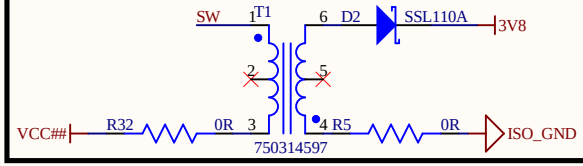
3.8-5V CHARGE PUMP



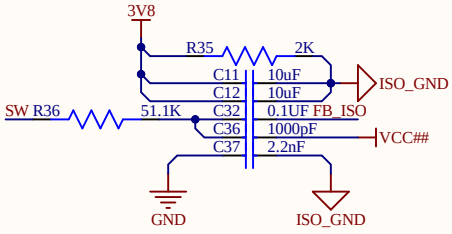
FEEDBACK NETWORK



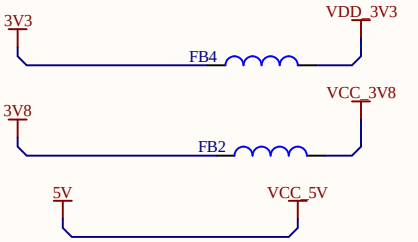
FLYBACK TRANSFORMER

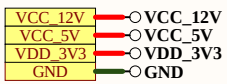


SUPPORTING CIRCUIT

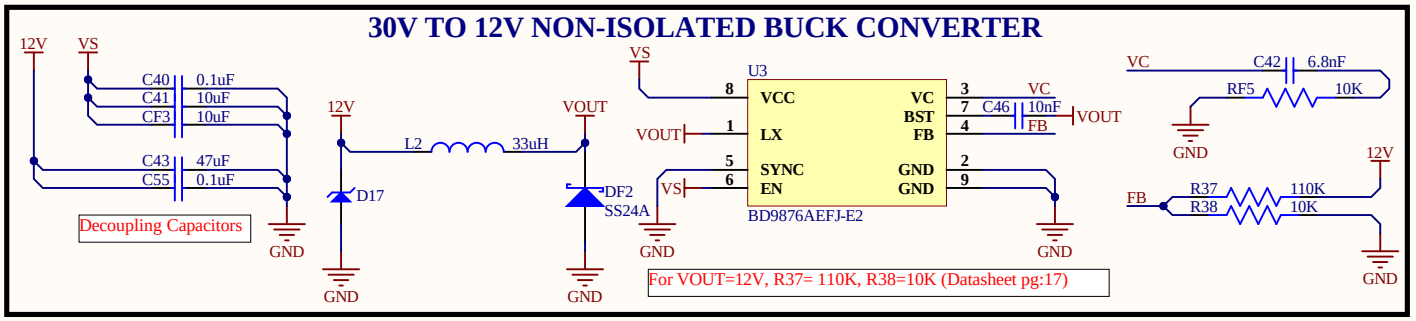


FERRITE CONNECTIONS

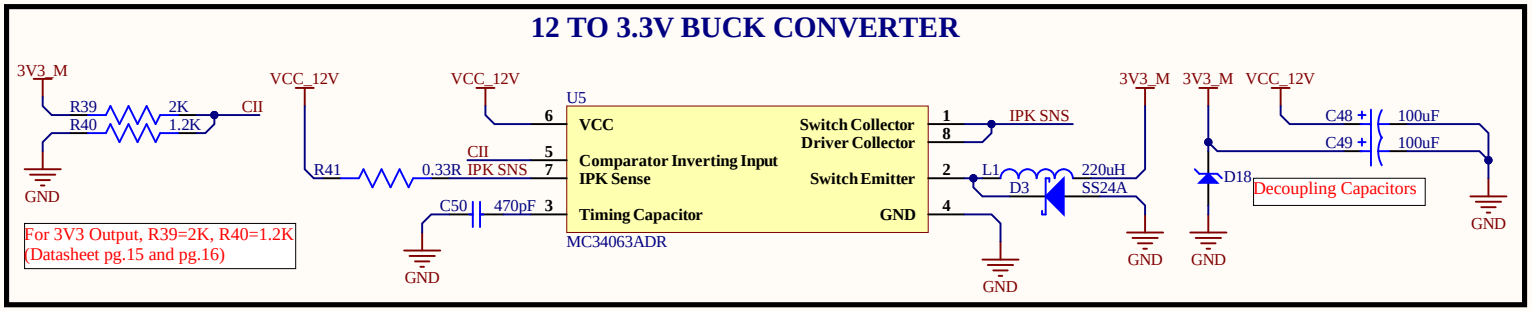




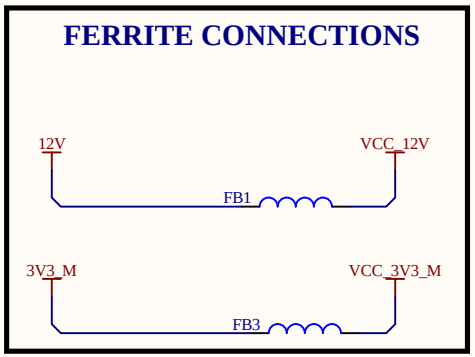
30V TO 12V NON-ISOLATED BUCK CONVERTER

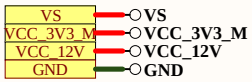


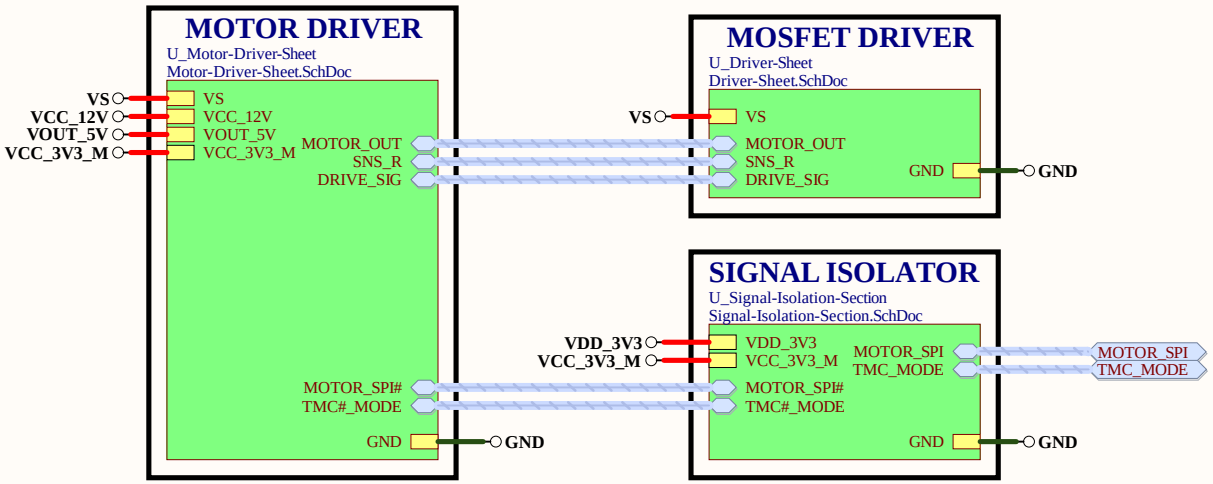
12 TO 3.3V BUCK CONVERTER

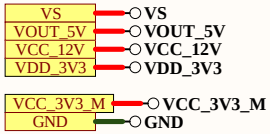


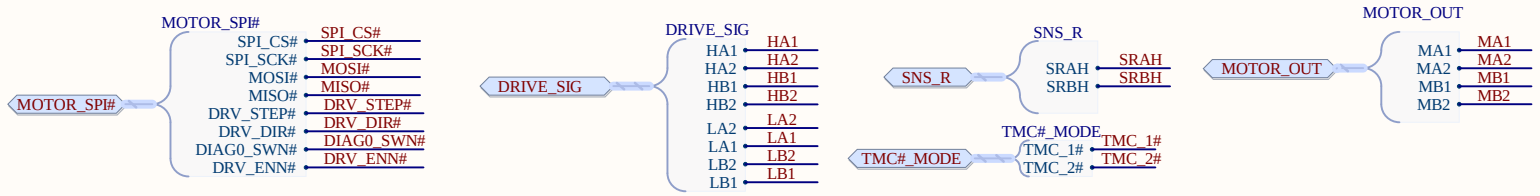
FERRITE CONNECTIONS



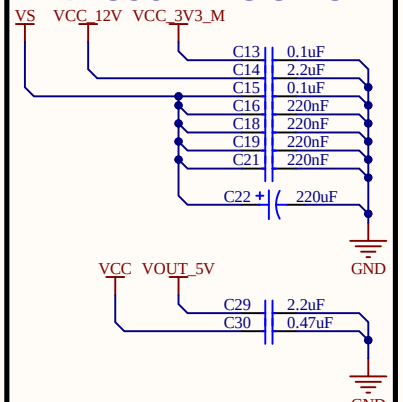



MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE


VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND

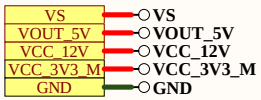


DECOUPLING CAPS

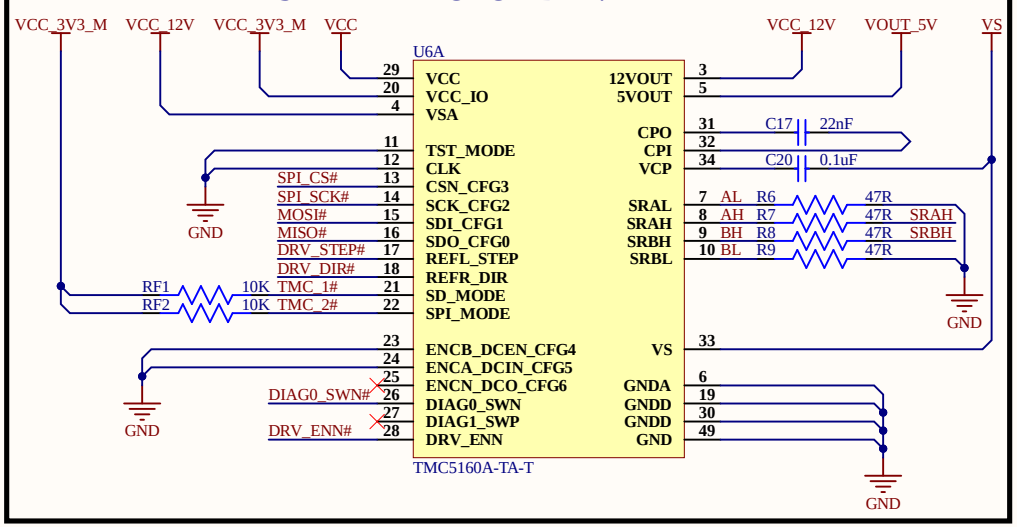


SERIES RESISTOR

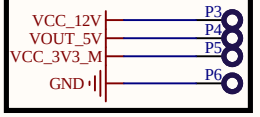




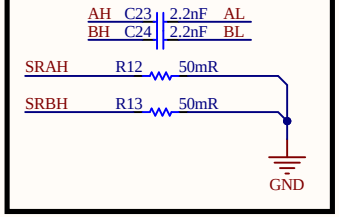
STEPPER MOTOR DRIVER-PART-A



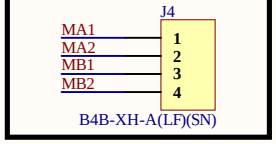
TEST POINTS



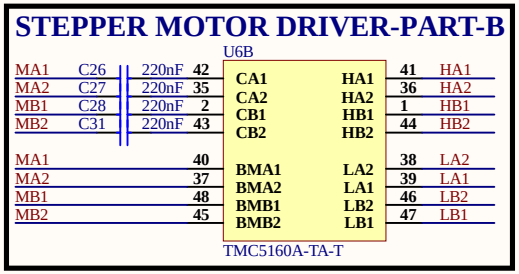
CURRENT SENSE RES



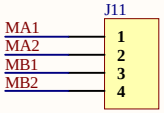
MOTOR OUTPUT

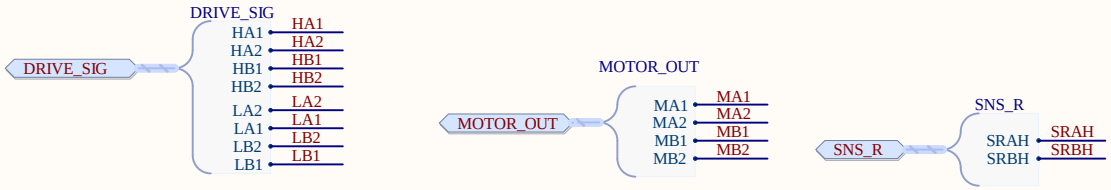


STEPPER MOTOR DRIVER-PART-B

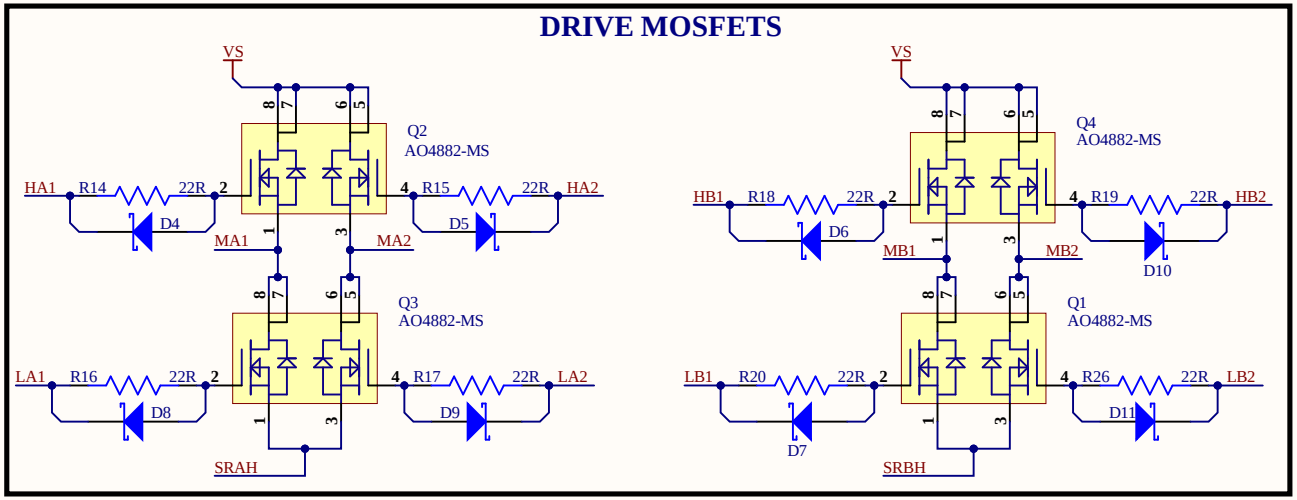


MOTOR OUTPUT FOR NEMA34

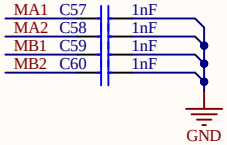




DRIVE MOSFETS



DECOUPLING CAPS





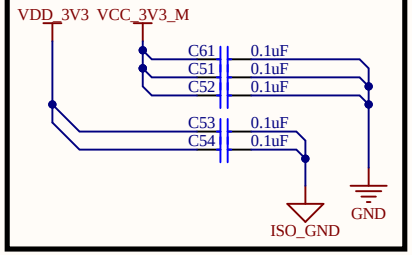
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN

MOTOR_SPI#
SPI_CS#
SPI_SCK#
MOSI#
MISO#
DRV_STEP#
DRV_DIR#
DIAG0_SWN#
DRV_ENN#

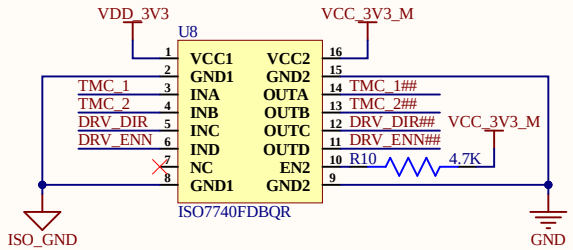
TMC_MODE
TMC_1
TMC_2

TMC#_MODE
TMC_1#
TMC_2#

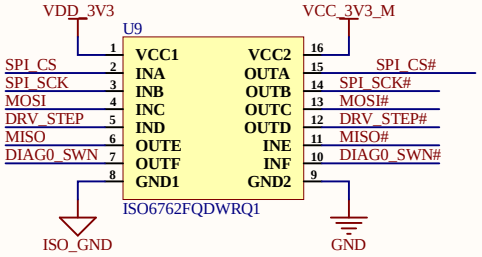
DECOUPLING CAPS



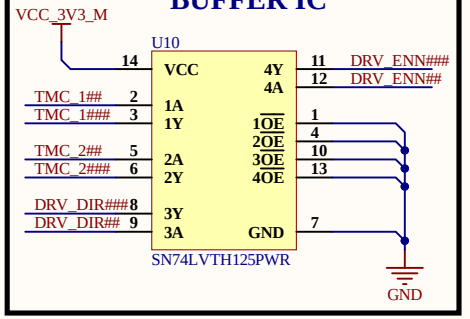
UNIDIRETIONAL SIGNAL ISOLATOR



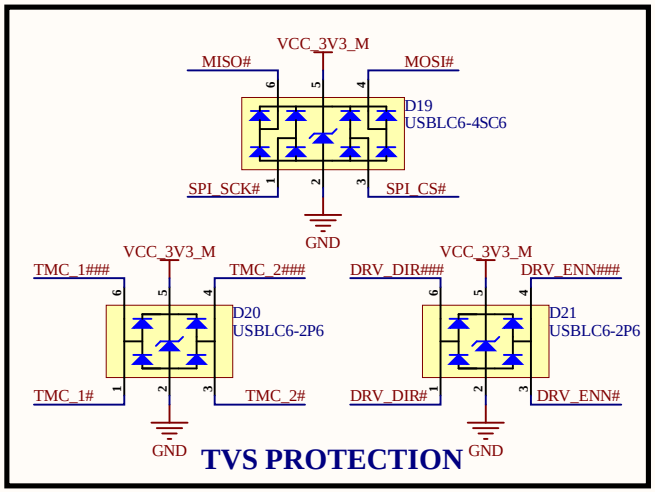
UNIDIRECTIONAL SIGNAL ISOLATOR



BUFFER IC



TVS PROTECTION

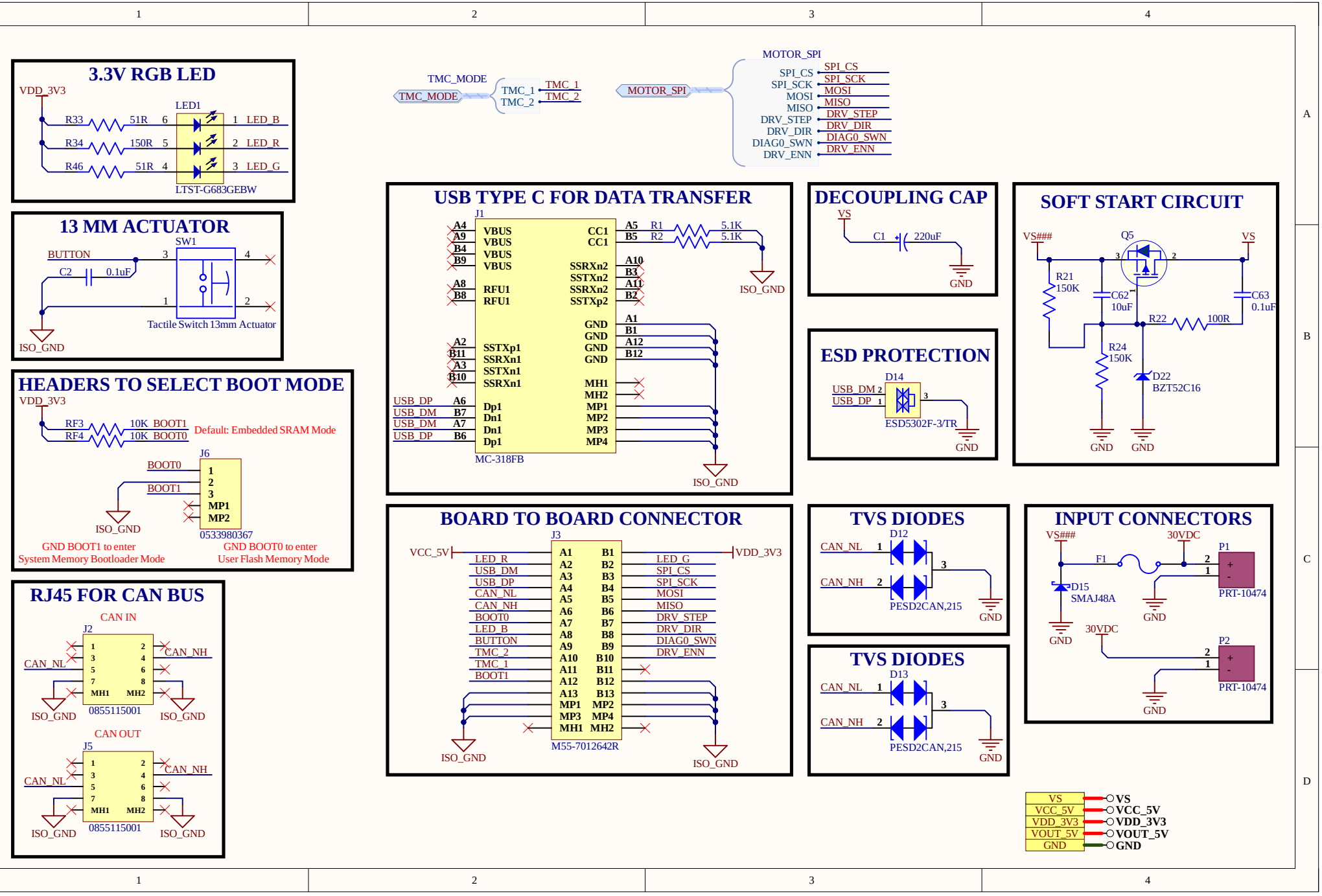


VDD_3V3 — VDD_3V3

VCC_3V3_M — VCC_3V3_M

GND — GND

Title		
Size A4	Number	Revision
Date: 1/10/2024		Sheet of
File: C:\Users\..\Signal-Isolation-Section.SchDoc		Drawn By:


3.3V RGB LED
VDD_3V3
LED1
R33 51R 6 1 LED_B
R34 150R 5 2 LED_R
R46 51R 4 3 LED_G
LTST-G683GEBW
13 MM ACTUATOR
SW1
BUTTON
C2 0.1uF
Tactile Switch 13mm Actuator
ISO_GND
HEADERS TO SELECT BOOT MODE
VDD_3V3
RF3 10K BOOT1
RF4 10K BOOT0
Default: Embedded SRAM Mode
J6
BOOT0
BOOT1
MP1
MP2
0533980367
GND BOOT1 to enter System Memory Bootloader Mode
GND BOOT0 to enter User Flash Memory Mode
RJ45 FOR CAN BUS
CAN IN
J2
CAN_NL
CAN_NH
0855115001
CAN OUT
J5
TMC_MODE
TMC_1
TMC_2
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
USB TYPE C FOR DATA TRANSFER
J1
VBUS
RFU1
SSTXp1
SSRXn1
SSTXn1
SSRXn1
Dp1
Dn1
CC1
SSRXn2
SSTXn2
SSRXn2
SSTXp2
GND
MH1
MH2
MP1
MP2
MP3
MP4
R1 5.1K
R2 5.1K
MC-318FB
DECOUPLING CAP
VS
C1 220uF
GND
ESD PROTECTION
D14
ESD5302F-3/TR
SOFT START CIRCUIT
VS###
Q5
R21 150K
C62 10uF
R22 100R
C63 0.1uF
R24 150K
D22 BZT52C16
BOARD TO BOARD CONNECTOR
J3
VCC_5V
VDD_3V3
LED_R
LED_G
USB_DM
SPI_CS
USB_DP
SPI_SCK
CAN_NL
MOSI
CAN_NH
MISO
BOOT0
DRV_STEP
LED_B
DRV_DIR
BUTTON
DIAG0_SWN
TMC_2
DRV_ENN
TMC_1
BOOT1
M55-7012642R
TVS DIODES
D12
PESD2CAN,215
D13
INPUT CONNECTORS
30VDC
F1
P1
PRT-10474
D15 SMAJ48A
P2
VS
VCC_5V
VDD_3V3
VOUT_5V
GND